使用GPU切片和虚拟化技术优化AI工作负载

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1. 调度原因可视化

PR: https://github.com/Project-HAMi/HAMi/issues/1332

```
Events:
  Type     Reason          Age                    From           Message
  ----     ------          ----                   ----           -------
  Warning  FilteringFailed  12m (x15 over 5d2h)    hami-scheduler  5 nodes failed: CardTypeMismatch
(node1,node2,node3,node4,node5)
  Warning  FilteringFailed  93s (x8 over 12d)      hami-scheduler  3 nodes failed:
CardInsufficientMemory (node2,node3,node5)
  Warning  FilteringFailed  93s (x8 over 12d)      hami-scheduler  1 node failed:
ExclusiveDeviceAllocateConflict (node5)
  Warning  FilteringFailed  93s (x8 over 12d)      hami-scheduler  4 nodes failed:
CardComputeUnitsExhausted (node1,node2,node3,node4)
```

第一章：HAMi v2.7新增功能

2. 改进的resourceQuota

```
# HELP GPUDeviceMemoryAllocated Device memory allocated for a certain GPU
# TYPE GPUDeviceMemoryAllocated gauge
GPUDeviceMemoryAllocated{deviceidx="0",deviceuuid="GPU-6ba25656-d7fa-c0f5-aa41-647d492d23fb",nodeid="k8s-node02",zone
# HELP GPUDeviceMemoryLimit Device memory limit for a certain GPU
# TYPE GPUDeviceMemoryLimit gauge
GPUDeviceMemoryLimit{deviceidx="0",deviceuuid="GPU-6ba25656-d7fa-c0f5-aa41-647d492d23fb",nodeid="k8s-node02",zone="vG
# HELP GPUDeviceSharedNum Number of containers sharing this GPU
# TYPE GPUDeviceSharedNum gauge
GPUDeviceSharedNum{deviceidx="0",deviceuuid="GPU-6ba25656-d7fa-c0f5-aa41-647d492d23fb",nodeid="k8s-node02",zone="vGPU
# HELP QuotaUsed resourcequota usage for a certain device
# TYPE QuotaUsed gauge
QuotaUsed{limit="0",quotaName="nvidia.com/gpucores",quotanamespace="dev",zone="vGPU"} 0
QuotaUsed{limit="0",quotaName="nvidia.com/gpumem",quotanamespace="dev",zone="vGPU"} 2000
# HELP nodeGPUMemoryPercentage GPU Memory Allocated Percentage on a certain GPU
# TYPE nodeGPUMemoryPercentage gauge
nodeGPUMemoryPercentage{deviceidx="0",deviceuuid="GPU-6ba25656-d7fa-c0f5-aa41-647d492d23fb",nodeid="k8s-node02",zone=
# HELP nodeGPUOverview GPU overview on a certain node
# TYPE nodeGPUOverview gauge
nodeGPUOverview{deviceidx="0",devicememorylimit="7680",devicetype="NVIDIA-Tesla P4",deviceuuid="GPU-6ba25656-d7fa-c0f
```

第一章：HAMi v2.7新增功能

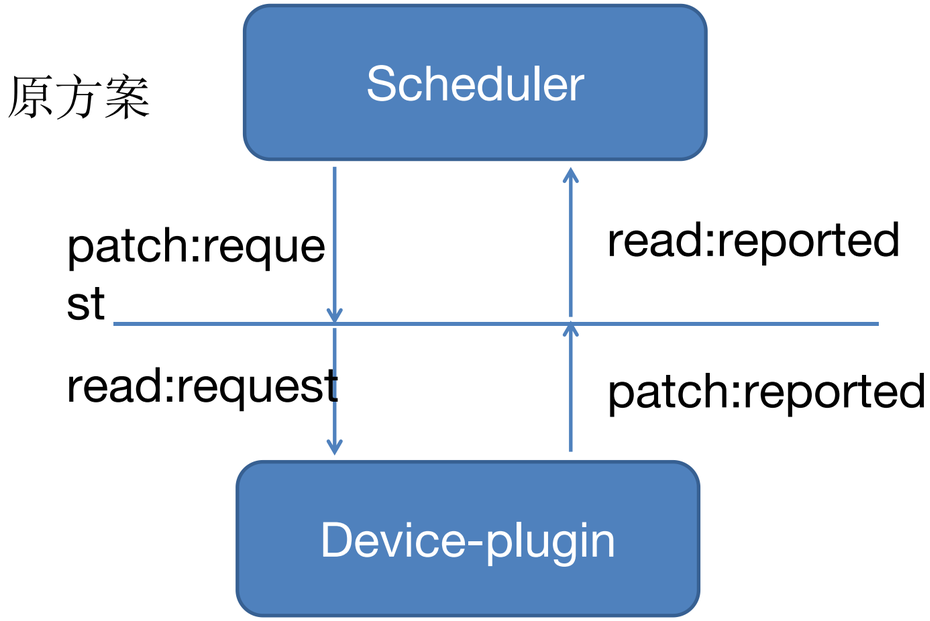
3. 支持AWS神经网络芯片/昆仑芯vxpu

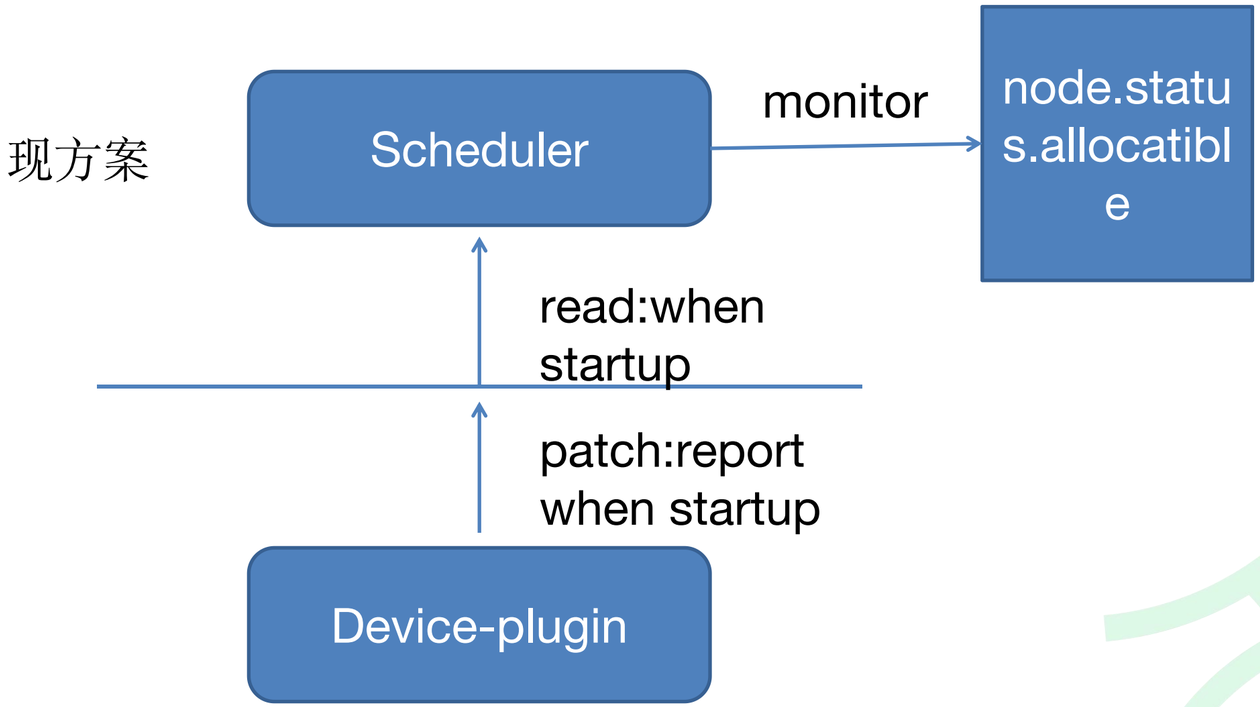
昆仑芯

aws

第二章：HAMi v2.8RoadMap

1. 资源上报优化





第二章：HAMi v2.8RoadMap

2. WebUI优化，支持异构设备

第二章：HAMi v2.8RoadMap

2. 支持的新设备

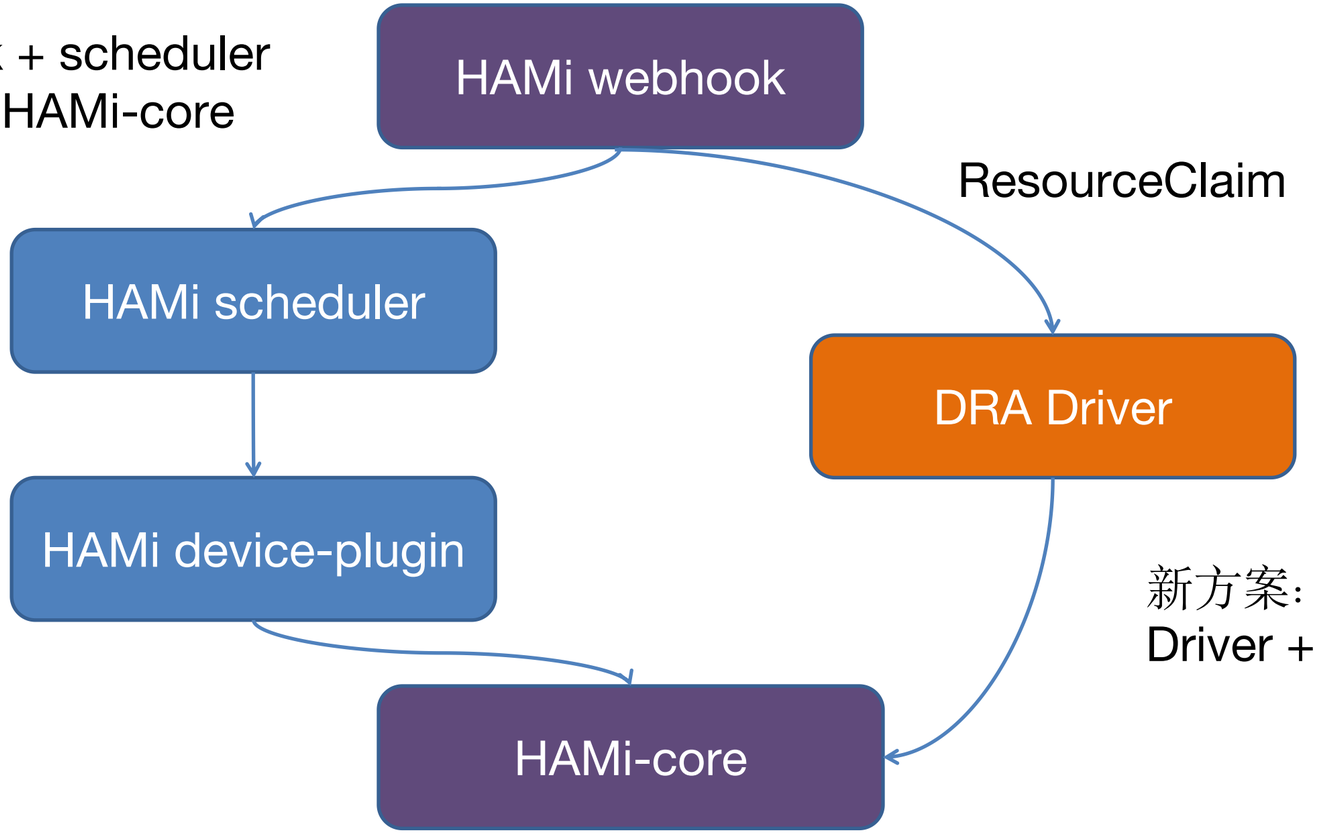
阿里云

AMD

第三章：HAMi DRA

方案设计

原方案：webhook + scheduler + device-plugin + HAMi-core



新方案：webhook + DRA Driver + HAMi-core

第三章：HAMi DRA

预期效果

功能名称	HAMi	HAMi DRA v1.0
动态显存/算力分配		
显存/算力多纬度分配		
资源硬隔离		
指定UUID		
指定设备类型		
资源配额		
调度器监控		
调度策略（binpack/spread)		
拓扑调度	完整	简单